COSMOLOGY

Problem Set 3: The Thermal History Of The Universe

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These are the end of chapter problems for Chapter 3 of the notes. Equation numbers refer to the full notes.

1. Gravitinos

In supergravity theories, the superpartner of the graviton, the gravitino, is a spin- 3/2 dark matter candidate. Assuming a negligible energy density of gravitinos after inflation, these particles would be produced most efficiently just after reheating. The rate at which Standard Model particles in the primordial plasma would convert into gravitinos is $\Gamma_g \sim T^3/M_{\rm P1}^2$. In this problem, we will estimate the relic density of gravitinos.

a). First, argue that the normalized number of gravitinos, $N_g \equiv n_g/s$, is given by

$$N_g \sim \int_0^{T_R} \frac{\mathrm{d}T}{T} \frac{\Gamma_g}{H} \frac{n_g}{s},$$

where T_R is the reheating temperature (temperature at the beginning of the universe), s is the total entropy density and n_g is the particle density. Using that $n_g/s \sim O(1)$ and $H \sim T^2/M_{\rm Pl}$, derive a simple expression for N_g in terms of T_R and $M_{\rm Pl}$.

b). Assuming one massless neutrino species, show that the total entropy density of the universe today is $s_0 \approx 2000\ \mathrm{cm}^{-3}$.

c). Show that the energy density of gravitinos today would be

$$\Omega_g h^2 \sim 0.1 \left(\frac{T_R}{10^9 \mathrm{GeV}}\right) \left(\frac{m_g}{1\mathrm{GeV}}\right).$$

For a reheating temperature of $T_R \sim 10^9$GeV, gravitinos with masses around $m_g \sim 1$GeV could hence constitute all of the dark matter. Note, however, that this scenario is severely constrained by its effects on BBN through the decay of heavier supersymmetric particles.

2. The Baryon-To-Photon Ratio

a). Show that the number density of CMB photons today is $n_{\gamma,0} \approx 411\ \mathrm{cm}^{-3}$.

b). Using $\Omega_b h^2 = 0.0224$ and $\rho_{\mathrm{crit},0} = 1.05 \times 10^{-5} h^2$ GeV cm^{-3}, compute the number

density of baryons today and the ratio $\eta = n_b/n_\gamma$. Why is η constant once $e^\pm$ annihilation is over?

3. The Neutrino Background

a). Using (3.61) and (3.30), compute the number density of each neutrino species today (counting neutrinos and antineutrinos).

b). If one neutrino species has a mass $m_\nu \gg T_{\nu,0}$, show that its contribution to the density today is $\Omega_\nu h^2 = m_\nu/(94\text{ eV})$. What does the observed $\Omega_m h^2 \approx 0.14$ say about the neutrino masses?

4. Helium As A Probe Of New Physics

Suppose there were an additional relativistic species in equilibrium at BBN, increasing $g_\star$ from 10.75 to 12.5. Qualitatively, how would this change the freeze-out temperature of the neutron-to-proton ratio, the time of nucleosynthesis, and the helium fraction Y_P? Estimate the change in T_f using $\Gamma \propto G_F^2 T^5$ and $H \propto g_\star^{1/2} T^2$.